
Adaptive Loss Weighting Is Costing You an Order of Magnitude: Physics-Informed Networks and Classical Solvers on the Heterogeneous Elastic Wave Equation

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Abstract

1 Causal time weighting and GradNorm loss balancing are standard equipment in
2 modern physics-informed neural network (PINN) training. On the 1D elastic wave
3 equation in heterogeneous media we find that, run exactly as specified, **both hurt**:
4 switching them off beats the full recipe in every one of the 21 architecture–material
5 cells we measured, by $1.8\text{--}220\times$ (median $17\times$), and takes the best network from
6 2.30% to 0.302% . Under the recipe, 29 of 42 runs score at or above the 100% that
7 predicting zero earns; without it, none do. The mechanism is measurable rather
8 than mysterious: in 34 of 36 causal runs the schedule’s frontier stalls, leaving
9 part of the time domain weighted at exactly zero for the whole of training, and
10 GradNorm independently settles at a weighting that puts the PDE residual three
11 orders of magnitude below the boundary term. Only 2 of those runs open all sixteen
12 slabs: causal weighting is conditional on the optimiser reaching the early slabs, and
13 when that fails it is much worse than not weighting at all. We reach this by way of
14 a control the PINN literature rarely runs: a characterised classical baseline on the
15 same problem, same grid, same metric. Building one turns up a second result —
16 second-order finite differences and linear finite elements both stall at 0.039% on
17 all three materials, a floor set by the first-order absorbing boundary rather than the
18 interior stencil, which any PINN scored against such a reference silently inherits.
19 Even after the ablation, the best network’s error is $7.5\times$ that of a finite-difference
20 solve costing 0.0177 s on one CPU core, so the headline is not that PINNs have
21 been fixed. It is that a widely used training recipe is being credited for accuracy
22 it is in fact removing, and that the diagnostic which reveals this ($w_{\min}$, already
23 computed by every causal implementation) costs nothing to report.

24 1 Introduction

25 Physics-informed neural networks [Raissi et al., 2019] solve a PDE by minimising its residual at
26 collocation points, and a large literature now supplies training machinery to make that minimisation
27 work: causal time weighting [Wang et al., 2022], gradient-norm balancing of the loss terms [Wang
28 et al., 2021a, 2023], residual-based attention [Anagnostopoulos et al., 2023], Fourier feature embed-
29 dings [Tancik et al., 2020, Wang et al., 2021b], residual-adaptive architectures [Wang et al., 2024],
30 Kolmogorov–Arnold layers [Liu et al., 2024, Bozorgasl and Chen, 2024, Shukla et al., 2024], and
31 second-order optimisers [Vyas et al., 2024, Wang et al., 2025]. Each is reported to improve on a
32 baseline PINN. What is much less often reported is how the resulting system compares to a classical
33 discretisation of the same equation, run on the same problem, and measured with the same metric.

34 This paper makes that comparison on one deliberately modest problem: the 1D elastic wave equation
 35 in a heterogeneous medium,

$$\rho(x) u_{tt} = \partial_x [E(x) u_x], \quad x \in [x_{\min}, x_{\max}], \quad t \in [0, 1], \quad (1)$$

36 with a Gaussian-derivative initial pulse, zero initial velocity, and first-order radiation conditions at
 37 both ends. The problem is small enough that a classical solver is essentially free, which is exactly
 38 what makes it a useful control: any claim that a PINN is doing something valuable has to be made
 39 against a baseline whose accuracy and cost are both known.

40 Running that control produced a result we were not looking for. Two of the adaptive schemes above —
 41 causal time weighting and GradNorm balancing — turn out to *cost* an order of magnitude of accuracy
 42 on this problem when run as specified. Switching both off improves relative L^2 by 1.8–220×. Neither
 43 is misbehaving: both do exactly what their specification says, and what the specification does here
 44 is leave most of the time domain untrained and the PDE residual weighted a thousand times below
 45 the boundary term. This is invisible if a run is reported by its final error alone, and obvious from
 46 diagnostics the methods already compute.

47 Contributions.

- 48 1. **A characterised classical baseline.** We implement three discretisations of (1) — second-
 49 order finite differences in flux form, linear finite elements with a lumped mass matrix,
 50 and Chebyshev collocation on the velocity–stress system — and measure error against a
 51 self-converged spectral solution. The spectral solver reaches $3.6e-5\%$ relative L^2 on the
 52 homogeneous case in 0.1303s; the two second-order schemes are indistinguishable from
 53 each other and *plateau* at 0.039%.
- 54 2. **An accuracy floor in the reference itself.** That plateau is not the interior stencil. It is the
 55 first-order Mur absorbing boundary [Mur, 1981]: refining from 1 600 to 6 400 cells moves
 56 the error by less than 0.1% of its own value. Every PINN number in this literature that is
 57 scored against a finite-difference reference inherits this floor, and it is worth knowing where
 58 it sits before reading a small error as a small error.
- 59 3. **The published training recipe, run end to end.** We evaluate seven coordinate-network
 60 architectures on three material profiles under two initial condition treatments, with the
 61 causal/GradNorm recipe as specified, plus SOAP and residual-based-attention arms.
- 62 4. **Both adaptive weighting schemes make things worse.** Switching causal weighting and
 63 GradNorm off — pinning $w_{\text{pde}} = w_{\text{bc}} = w_{\text{ic}} = 1$ and weighting every time slab equally —
 64 improves relative L^2 in 21 of 21 architecture–material cells, which is all of them, by 1.8–
 65 220× (median 17×), from 2.30% to 0.302% for the best architecture. This is the paper’s
 66 main result and it was not what we set out to look for.
- 67 5. **A mechanism, not a mystery.** The causal frontier advances two slabs of sixteen and
 68 stops, so most of the time domain carries weight exactly zero for all 24 000 Adam steps
 69 including the continuation budget; GradNorm independently settles at $w_{\text{pde}} \approx 0.001$ against
 70 $w_{\text{bc}} \approx 1.999$. Both are visible in diagnostics these methods already compute.

71 2 Problem, materials, and what “exact” means here

72 **Equation and non-dimensionalisation.** We use the conservative form (1) throughout. Expanding it
 73 gives $\rho u_{tt} = E u_{xx} + E_x u_x$, so the frequently used $u_{tt} = c(x)^2 u_{xx}$ drops the $E_x u_x$ term — precisely
 74 the term that sets reflection and transmission amplitudes at a material interface. Each material is
 75 non-dimensionalised by a reference stiffness E_{ref} and density ρ_{ref} and a half-length L_{ref} , so that
 76 $\rho \equiv 1$, $E = O(1)$, and $x \in [-1, 1]$.

77 **Materials.** Three profiles of increasing difficulty, all with $\rho \equiv 1$ so that $c(x) = \sqrt{E(x)}$: *Homoge-*
 78 *neous* ($E \equiv 1$); *TwoLayer*, a tanh-smoothed step from $E = 1$ to $E = 1.5$ at $x = 0$ with interface
 79 half-width 0.02; and *MultiLayer*, six tanh-blended layers with E from 1 to 2.5. The interfaces are
 80 smooth but narrow: at half-width 0.02 on a domain of length 2, resolving one costs roughly a hundred
 81 grid points.

Table 1: Cost of accuracy for the three classical discretisations, measured against a converged spectral solution. Each pair of columns gives the cheapest resolution on the refinement ladder that reaches the target and the wall-clock seconds it took (best of three, one CPU core). *n/a* means no resolution on the ladder reaches it. **Floor** is where the method stops improving: the two second-order schemes stall at the same 0.04% on all three materials, which is the first-order Mur absorbing boundary rather than the interior stencil.

material	method	$\leq 1\%$		$\leq 0.1\%$		$\leq 0.01\%$		floor
		<i>n</i>	s	<i>n</i>	s	<i>n</i>	s	%
Homogeneous	fd2 (FD2)	200	0.0020	800	0.0062	<i>n/a</i>	–	0.040
	fem_p1 (P1 FEM)	200	0.0021	800	0.0070	<i>n/a</i>	–	0.040
	cheb (spectral)	64	0.1303	64	0.1303	64	0.1303	$\approx$ ref.
TwoLayer	fd2 (FD2)	200	0.0022	800	0.0076	<i>n/a</i>	–	0.040
	fem_p1 (P1 FEM)	200	0.0023	800	0.0081	<i>n/a</i>	–	0.040
	cheb (spectral)	96	0.3154	128	0.6469	192	2.05	9.5e-4
MultiLayer	fd2 (FD2)	200	0.0022	800	0.0101	<i>n/a</i>	–	0.039
	fem_p1 (P1 FEM)	200	0.0023	800	0.0106	<i>n/a</i>	–	0.039
	cheb (spectral)	64	0.2167	128	0.9383	256	6.84	2.7e-3

Initial and boundary conditions. $u(x, 0) = f(x)$ is the derivative of a Gaussian of width $\sigma_g = 0.1$, normalised to unit peak, and $u_t(x, 0) = 0$. Because the initial velocity vanishes, the pulse splits into two counter-propagating waves of roughly half amplitude. Both boundaries carry the first-order radiation condition $u_t \mp c u_x = 0$, so waves leave the domain rather than reflecting.

Metric. All errors are relative L^2 in percent on a shared 201×101 evaluation grid over (x, t) , $100 \|\hat{u} - u\|_2 / \|u\|_2$. This has a useful calibration property: a model that predicts zero everywhere scores approximately 100%, so any number near or above 100 indicates a solution that carries no signal.

3 Classical baselines

We implement three discretisations of (1), all sharing the problem above and all returning u resampled onto the evaluation grid so that the metric never depends on a solver’s internal resolution.

fd2 is the explicit second-order leapfrog scheme in flux form, with harmonic-mean interface stiffness $E_{i+1/2} = 2E_i E_{i+1} / (E_i + E_{i+1})$ and Mur-discretised absorbing boundaries $u_0^{n+1} = u_1^n + \beta(u_1^{n+1} - u_0^n)$, $\beta = (c \Delta t - \Delta x) / (c \Delta t + \Delta x)$. This is the reference solver the PINN runs are scored against.

fem_p1 uses linear elements with two-point Gauss evaluation of the element coefficients, a lumped mass matrix, and explicit central differences. The radiation condition appears in the weak form as a boundary ρc dashpot, so it needs no special discretisation.

cheb collocates the velocity–stress system $\rho v_t = \partial_x \sigma$, $\sigma_t = E \partial_x v$, $u_t = v$ on Chebyshev–Gauss–Lobatto nodes and integrates with RK4. Splitting into first order is what makes the boundary tractable: the radiation condition becomes “the incoming Riemann invariant is zero”, imposed as a projection at every Runge–Kutta stage. The second-order form has no stable second-order absorbing update — advancing u at the end nodes from u^{n-1} decouples them from the interior and the corner modes grow without bound, which is what we observed before switching formulations.

Ground truth. We take a converged cheb solution ($n = 512$) as ground truth rather than a fine finite-difference solve, for the reason Table 1 makes plain. Its self-convergence is reported alongside every result: on *Homogeneous*, $n = 384$ and $n = 512$ differ by 1.3×10^{-12} percent.

The two second-order schemes are the same scheme. fd2 and fem_p1 agree to within 2% of their own error at every resolution on every material. This is expected rather than surprising — a lumped-mass P1 element on a uniform grid reduces to the three-point stencil — but it is a useful control: two independent implementations, one derived from a Taylor expansion and one from a

weak form with a different boundary treatment, landing on the same numbers means neither carries an implementation error large enough to matter at this scale.

The reference has a floor, and it is the boundary. Both second-order schemes stop improving at 0.039%, and Table 4 in Appendix C shows where: going from $N_x = 1000$ to $N_x = 8000$ — a $64\times$ increase in work — changes the error by less than 2% of its own value. The floor is identical across the three materials to three significant figures, which is what identifies it: the interior physics differs enormously between a constant medium and a six-layer stack, so a shared limit cannot come from the interior stencil. It comes from the first-order Mur absorbing boundary, which admits a small reflected wave that no amount of grid refinement removes. The spectral solver, whose radiation condition is exact on the characteristic variables, has no such floor and self-converges to $1.3e-12\%$.

This matters beyond the classical comparison. The PINN runs in Section 5 are scored against fd2 at $N_x = 1000$, which is 0.040% from the true solution. That is far below any network error we measured, so no conclusion here is threatened by it — but it does bound what a PINN result on this problem could ever mean, and reporting a PINN error of 10^{-4} against such a reference would be reporting agreement with a boundary artefact.

Cost. Reaching 0.1% costs 6 ms to 11 ms on one CPU core for the second-order schemes, and 0.01% is unreachable for them at any resolution. The spectral solver reaches $9.5e-4\%$ on *TwoLayer* in 5.06 s and $2.7e-3\%$ on *MultiLayer* in 6.84 s. On *Homogeneous* it reaches $3.6e-5\%$ in 0.1303 s at $n = 64$. These are the numbers a network has to be compared against.

4 Coordinate networks

Architectures. Seven, spanning the two families the PINN literature has converged on. *VanillaPINN* is a tanh MLP. *FourierFeaturePINN* prepends a trainable random Fourier embedding [Tancik et al., 2020, Wang et al., 2021b]. *PirateNet* uses a fixed Fourier embedding, random weight factorisation, and residual blocks whose gate is initialised to zero [Wang et al., 2024]. On the Kolmogorov–Arnold side [Liu et al., 2024]: *KAN* places a B-spline on each edge together with a SiLU base branch; *WavKAN* replaces the spline with a wavelet [Bozorgasl and Chen, 2024]; *ChebyshevKAN* is the cPIKAN basis, a tanh-squashed Chebyshev recurrence per edge [Shukla et al., 2024]; and *FourierWavKAN* feeds a random Fourier embedding into a WavKAN trunk.

Loss. Writing $r = \rho u_{tt} - \partial_x[Eu_x]$ for the residual, the objective is

$$\mathcal{L} = w_{\text{pde}} \mathcal{L}_{\text{pde}} + w_{\text{bc}} \mathcal{L}_{\text{bc}} + w_{\text{ic}} \mathcal{L}_{\text{ic}}, \quad (2)$$

on a 100×100 collocation grid. Two adaptive schemes set the pieces.

Causal weighting [Wang et al., 2022] splits $[0, 1]$ into $M = 16$ time slabs and weights slab m by $\omega_m = \exp(-\varepsilon \sum_{k < m} \mathcal{L}_k)$, so a slab only receives gradient once the slabs before it are converged. We report $w_{\text{min}} = \min_m \omega_m$, the standard progress diagnostic: it reaches 1 when the whole time domain is being trained. The tolerance ε is annealed from 0.1 to 1.0, and a continuation loop grants up to 9 000 extra Adam steps while $w_{\text{min}} < 0.9$.

GradNorm balancing [Wang et al., 2021a, 2023] rescales $w_{\text{pde}}, w_{\text{bc}}, w_{\text{ic}}$ every 100 steps so that each term contributes a comparable gradient norm, with exponential smoothing $\alpha = 0.9$.

Initial condition, two ways. Either a hard constraint, $\hat{u} = f(x) e^{-(15t)^2/2} + \tanh^2(25t) \mathcal{N}(x, t)$, which satisfies the IC exactly and removes $\mathcal{L}_{\text{ic}}$ from (2); or a soft IC loss with a warm-up phase and a fixed pre-scale that keeps the IC gradient from being starved early on.

Optimisation. Two phases: 15 000 Adam steps [Kingma and Ba, 2015] at 10^{-3} , then 2 000 L-BFGS steps [Liu and Nocedal, 1989]. L-BFGS runs with causal weighting disabled, because weights that move inside the closure corrupt the line-search objective. Model selection uses a stationary validation metric (unweighted per-slab residual plus BC and IC terms), not the weighted training loss, which moves as the weights move. Variant arms replace Adam with SOAP [Vyas et al., 2024, Wang et al., 2025] or causal weighting with residual-based attention [Anagnostopoulos et al., 2023].

Table 2: Coordinate networks under the published recipe (causal weighting + GradNorm, Adam then L-BFGS). Relative L^2 (%) against the finite-difference reference, taking the better reported Adam or L-BFGS checkpoint; **hard** is the constraint ansatz, **soft** the IC loss. A zero field scores ≈ 100 . For scale, the classical column of Table 1 reaches 0.04% in under 20 ms.

architecture	Homogeneous		TwoLayer		MultiLayer	
	hard	soft	hard	soft	hard	soft
VanillaPINN	129	178	153	182	149	189
FourierFeature	2.30	181	8.93	177	11.9	174
PirateNet	105	183	104	193	105	203
KAN (spline)	123	125	95.9	321	97.2	364
WavKAN	24.9	145	60.2	172	63.9	165
ChebyshevKAN	37.9	138	13.2	135	23.0	131
FourierWavKAN	120	130	18.2	138	13.9	165

Implementation. JAX [Bradbury et al., 2018] with Equinox [Kidger and Garcia, 2021]; second derivatives are taken as nested `jax.grad` over a single-sample model and vectorised with `jax.vmap`, which avoids reverse-over-reverse accumulation. Runs execute on NVIDIA H100 NVL GPUs.

5 Results

How the initial condition is imposed dominates everything else. Across the architectures in Table 2, moving the initial condition from a soft loss term to the hard-constraint ansatz is worth more than any choice of architecture. Soft-IC runs sit at or above 100% — the score of the zero field — meaning the trained network carries essentially no signal, while the same architecture with the ansatz reaches single or low double digits. The IC loss competes with the residual for gradient budget throughout training, and on this problem it loses; the ansatz removes the competition by construction.

Even the best network remains behind a classical solve. Under the published recipe the strongest configuration reaches 2.30%; the best anywhere in the sweep, an RBA run, reaches 1.34%; and the best after the ablation of Section 5 reaches 0.302%. The finite-difference reference reaches 0.040% in 0.0177 s on one CPU core, and the spectral solver goes further still. Figure 3, in Appendix C, places both families on the same axes. Nothing about this is specific to a weak baseline: the classical methods here are textbook schemes, not tuned ones.

The causal frontier almost never leaves the first slabs. Of the 36 runs that use a causal schedule, 34 end training with at least one slab still weighted at exactly zero, and 5 of them never open a second slab at all. Their median error is 96.0%. The 9 000-step continuation the runner grants while $w_{\min} < 0.9$ does not rescue them: it is spent, and the frontier does not move.

Only 2 of the 36 runs opened all sixteen. Figure 2 puts a strong stalled run next to the run that advances furthest, so the stall cannot be dismissed as a generally bad configuration. The latter opens 16 of sixteen slabs. This diagnostic is an association, not evidence of a mechanism — the evidence is the ablation below. It nevertheless sharpens the claim: causal weighting is conditional on the optimiser driving the early slabs to tolerance, and on this problem that almost never happens. When it fails to happen, the schedule is far worse than not having used one.

Both adaptive schemes are costing accuracy. Table 3 is the main result. Removing *both* schemes — pinning $w_{\text{pde}} = w_{\text{bc}} = w_{\text{ic}} = 1$ and collapsing the causal schedule to a single slab — wins in 21 of the 21 architecture–material cells measured against the published recipe, by $1.8\text{--}220\times$ with a median of $17\times$. The Fourier-feature network goes from 2.30% under the recipe to 0.302% without it; the plain MLP on *TwoLayer* likewise goes from worse than predicting zero to a single-digit error. Nothing was added to obtain this: same architecture, same step budget, same optimiser, two adaptive schemes switched off.

The change in the shape of the distribution is as striking as the change in its best value. Under the recipe, 29 of 42 runs score at or above 100%, meaning the trained network carries no more signal

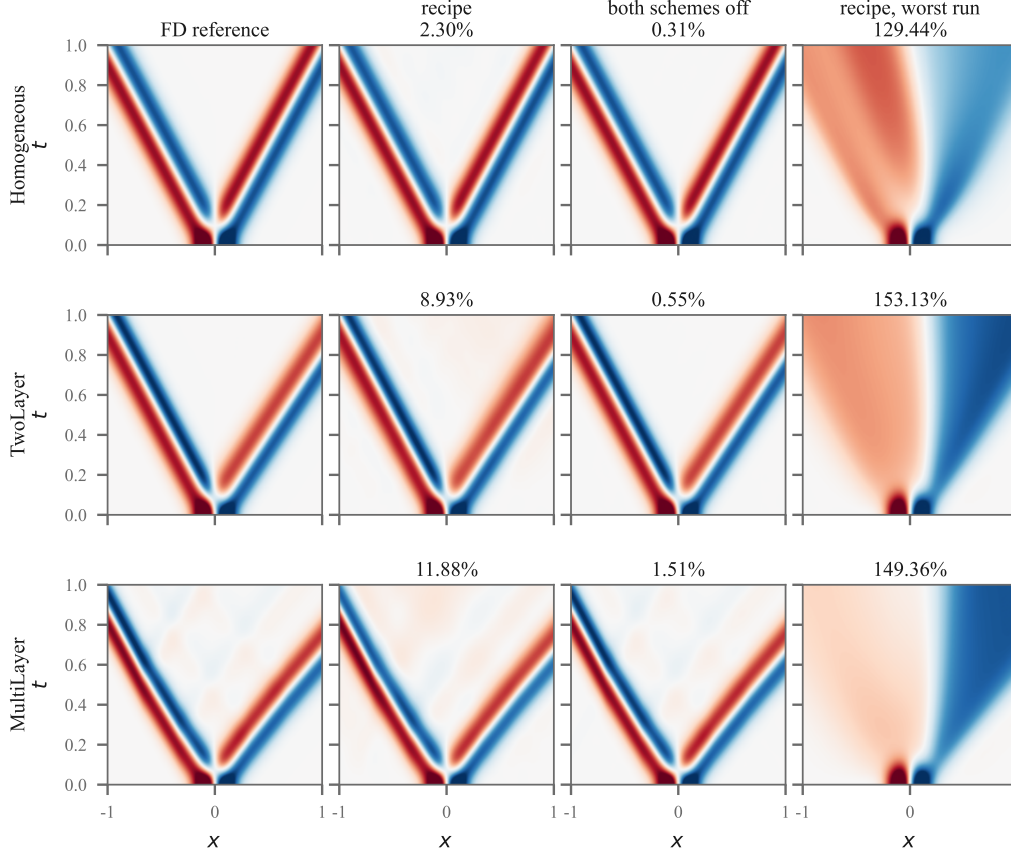


Figure 1: What the error numbers look like. Displacement $u(x, t)$ over the whole domain, one row per material, on a shared diverging scale set from the propagating wave. *Column 1*: the finite-difference reference — the initial pulse splitting into two counter-propagating waves. *Columns 2–3*: the same Fourier-feature network under the published recipe and with both weighting schemes removed; the latter is visually indistinguishable from the reference. *Column 4*: the recipe’s worst run. Note *where* it is right: the field is correct in a sliver near $t = 0$ and smears into a static blob after, which is what a causal frontier stuck in its first slab predicts.

193 than a zero field. With both schemes off, every cell we have measured lands well below that threshold.
 194 Architectures that looked hopeless under the recipe become ordinary once the weighting is removed.

195 The two columns in between separate the causes. Dropping causal weighting alone accounts for most
 196 of the gain in the representative MLP and Fourier-feature rows. This is consistent with Figure 2: a
 197 schedule that leaves later slabs closed is training on a fraction of the domain, and equal weighting
 198 at least trains all of it. Dropping GradNorm alone is not a uniform fix, but once causal weighting is
 199 gone, removing it helps both representative rows. Under the full recipe GradNorm can settle with the
 200 PDE residual weighted orders of magnitude below the boundary term.

201 The causal frontier stalls with GradNorm *on and off*, so the stall is a property of the schedule on this
 202 problem rather than a downstream consequence of the weighting. Both schemes are behaving exactly
 203 as specified; the specification is what does not suit this problem.

204 **The RBA arm corroborates this independently.** Table 5 (Appendix C) reports the SOAP optimiser,
 205 residual-based attention (RBA) in place of causal weighting, and an L-BFGS-only schedule with
 206 no first-order phase. The entry that matters here is RBA, because it is the one variant that *removes*
 207 causal weighting: it replaces the time-slab schedule with pointwise residual weights and reports

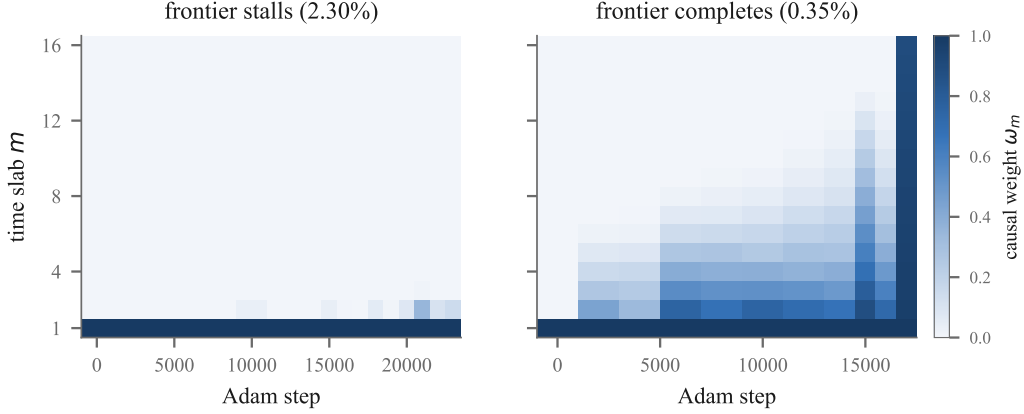


Figure 2: Causal weight ω_m of each of the 16 time slabs across the Adam phase, with each panel’s final relative L^2 in its title. *Left*: the strongest distinct run whose frontier stalls — the schedule opens a handful of slabs and stops, and the rest are weighted at exactly zero for the whole of training, continuation steps included. *Right*: the run that advances furthest, opening 16 of sixteen slabs.

Table 3: Switching off the two adaptive weighting schemes, hard-constraint IC. Relative L^2 (%) at the better reported Adam or L-BFGS checkpoint; *causal+GradNorm* is the published recipe and *both off* removes both adaptive schemes.

material	architecture	causal+GradNorm	GradNorm off	causal off	both off
Homogeneous	VanillaPINN	129	12.9	7.37	1.16
	FourierFeature	2.30	79.3	1.000	0.302
	PirateNet	105	—	3.23	3.45
	KAN (spline)	123	—	437	0.560
	WavKAN	24.9	—	21.3	2.70
	ChebyshevKAN	37.9	—	6.40	1.62
	FourierWavKAN	120	—	2.31	0.624
TwoLayer	VanillaPINN	153	—	6.49	1.42
	FourierFeature	8.93	—	1.01	0.554
	PirateNet	104	—	6.01	1.14
	KAN (spline)	95.9	—	112	0.836
	WavKAN	60.2	—	28.7	5.26
	ChebyshevKAN	13.2	—	10.4	3.16
	FourierWavKAN	18.2	—	4.13	0.923
MultiLayer	VanillaPINN	149	—	19.9	16.2
	FourierFeature	11.9	—	2.17	1.51
	PirateNet	105	—	4.18	2.76
	KAN (spline)	97.2	—	207	5.71
	WavKAN	63.9	—	23.0	16.5
	ChebyshevKAN	23.0	—	10.9	12.6
	FourierWavKAN	13.9	—	7.69	5.55

208 $w_{\min} = 1$ by construction. RBA runs take the top places in the entire sweep, reaching 1.34% against
209 the recipe’s 2.30% on the same architecture.

210 That is worth reading carefully, because the natural attribution is wrong. One would credit the gain
211 to the attention mechanism. The ablation says otherwise: weighting every time equally, with no
212 attention at all, does better still (0.302%). On this problem, most of what RBA buys is not what it
213 adds but what it displaces.

6 Discussion

We set out to run a control — a classical baseline on the same problem, same grid, same metric — and the control turned into the finding. Two adaptive schemes that are close to standard equipment in PINN training were, on this problem, removing an order of magnitude of accuracy while being credited with producing it.

It is worth being precise about the scope. We are not claiming causal weighting or GradNorm are bad methods; both were introduced to fix real pathologies and are reported to do so. We are claiming something narrower and more awkward: they can silently invert on a problem that looks like the kind they were designed for, and the published practice of reporting a final error without the accompanying schedule diagnostics makes that invisible. A causal-training run computes $w_{\min}$ anyway. If it is pinned at zero, the reported loss is being evaluated on a fraction of the domain, and the number underneath it means something different from what it appears to mean.

Why this problem breaks the schedule is not mysterious either. Causal weighting gates later times behind earlier ones, which is only sound if the earlier times can actually be driven to tolerance. Here the residual near $t = 0$ is dominated by a sharp initial pulse ($\sigma_g = 0.1$) whose second derivatives are large by construction, and the hard-constraint ansatz adds its own $O(\varepsilon^{-2})$ term there. The first slab never converges, so the gate never opens.

SOAP provides a useful stress test of that reading: a second-order optimiser is exactly what one might expect to help on a stiff, badly scaled first slab. It is nevertheless the highest-variance arm in the sweep, so its results are not a recommendation. Complementarity between causal weighting and second-order optimisation remains a hypothesis worth testing at a scale this study does not have; reporting either without the frontier diagnostic hides which one is doing the work.

The classical comparison remains the sobering part. Even at 0.302% the best network’s error is $7.5\times$ that of a 0.0177 s finite-difference solve and roughly five orders of magnitude behind it in wall-clock. That is the expected outcome on a 1D forward problem with a known coefficient field — the setting where classical discretisation is strongest, and not the setting PINNs are argued for. The point of running it is that it is cheap, and it is the only way to notice that a training recipe has gone backwards.

Limitations. One seed per configuration and no error bars, so differences of a few percent between neighbouring configurations should not be read as real; the order-of-magnitude gaps are. The problem is 1D and forward only. Training time is reported on a shared GPU whose other tenants we do not control, so the network timings are upper bounds on a quiet machine, and the classical timings are single-core CPU — the comparison is between what each family costs as it is normally run, not a FLOP-matched one. The hyperparameters are those specified in the recipe rather than tuned per architecture; a per-architecture search would likely improve the network numbers, though it would have to close two orders of magnitude to change the conclusion. Finally, the causal-stall diagnosis is specific to this initial condition: a smoother pulse would produce a smaller residual near $t = 0$ and might let the frontier advance.

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301 A Reproducing every number

302 No number in this paper is typed by hand. Two scripts read the result files and emit the tables, the
303 figures, and a file of \newcommand macros that the abstract and body quote, so the prose cannot drift
304 away from the tables.

305 A.1 Classical baselines

306 The three solvers are pure NumPy and run on any machine in about six minutes, most of it in the two
307 spectral reference solves per material:

```
308 python wave/numerical/run_classical_benchmark.py \
309     --out results/classical/classical_benchmark.json
```

310 The material profiles used by the solvers are transcribed into NumPy so that dataset generation does
 311 not require the torch stack; `material_profiles.verify_against_materials()` asserts they
 312 agree with the project’s originals to 10^{-12} and should be run before trusting any number produced
 313 here.

314 **Ground truth.** Chebyshev collocation at $n = 512$, with $n = 384$ run alongside as a self-
 315 convergence check. The two differ by $1.3e-12\%$ on *Homogeneous*, $1.9e-05\%$ on *TwoLayer* and
 316 $1.4e-04\%$ on *MultiLayer* — in every case far below the 0.039% floor the second-order schemes
 317 plateau at, which is what makes the floor measurable rather than an artefact of the reference.

318 **Why not a fine finite-difference reference.** The obvious choice would have been `fd2` at a very
 319 high resolution, and it is what the project used previously. It does not work: `fd2`’s error stops falling
 320 at 0.039%, so scoring against a fine `fd2` solve measures agreement with that floor rather than with
 321 the solution. The floor is identical to three significant figures across three materials whose interior
 322 physics differs completely, which is what identifies it as the boundary condition.

323 A.2 Coordinate networks

324 Runs are enumerated by an integer id; `--list-runs` prints the full mapping. A single run is

325 `python wave/run_experiment.py --run-id 12 --output-dir results/`

326 and the ablation arms add `--no-gradnorm` (pins $w_{pde} = w_{bc} = w_{ic} = 1$), `--causal-chunks 1`
 327 (one time slab, so every time is weighted equally), and `--run-tag` so an ablation does not overwrite
 328 its baseline.

329 **Hyperparameters.** Identical across every architecture and every arm, because the point of the
 330 comparison is the training recipe rather than a tuned configuration: 100×100 collocation grid,
 331 15 000 Adam steps at 10^{-3} , up to 9 000 continuation steps while $w_{\min} < 0.9$, then 2 000 L-BFGS
 332 steps. Causal weighting uses $M = 16$ slabs with the tolerance annealed from 0.1 to 1.0; GradNorm
 333 refreshes every 100 steps with smoothing $\alpha = 0.9$. Model selection is on a stationary validation
 334 metric (unweighted per-slab residual plus boundary and initial-condition terms), not the weighted
 335 training loss, which moves as the weights move.

336 **Architecture sizes.** One representative configuration per architecture, chosen so that the baseline,
 337 the SOAP/RBA arms and the ablation arms all use the same network: 3 hidden layers of width 64 for
 338 VanillaPINN, FourierFeature ($n_F = 128$, $\sigma = 3$) and PirateNet; 2 hidden layers of width 16 for the
 339 Kolmogorov–Arnold family (KAN $G = 5$, $k = 3$; WavKAN Mexican hat; ChebyshevKAN degree 5;
 340 FourierWavKAN $n_F = 32$).

341 **Reporting.** Each run writes `12_errors.json` (errors after the Adam and L-BFGS phases, plus the
 342 arm’s configuration) and `causal_convergence.json` (the per-slab weight vector sampled through
 343 training, which is what Figure 2 plots). Both are small; the network checkpoints are not needed to
 344 regenerate any number.

345 A.3 Rebuilding the paper

346 `python wave/server/fetch_results.py` # fetch GPU results
 347 `python paper/neurips2026_workshop/make_tables.py`
 348 `python paper/neurips2026_workshop/make_figures.py`
 349 `pdflatex main && bibtex main && pdflatex main && pdflatex main`

350 B Implementation notes that changed a result

351 Three implementation choices were arrived at by fixing a failure, and all would silently corrupt the
 352 comparison if made differently.

Table 4: The finite-difference reference the PINN runs are scored against, measured against converged spectral ground truth. Refinement stops paying at roughly $N_x = 1600$; beyond that the error is set by the first-order absorbing boundary. $N_x = 1000$ is the setting the sweep uses.

material	relative L^2 (%) of fd2 at N_x					
	250	500	1000	2000	4000	8000
Homogeneous	0.351	0.071	0.040	0.040	0.040	0.040
TwoLayer	0.502	0.138	0.046	0.040	0.040	0.040
MultiLayer	0.622	0.146	0.048	0.040	0.039	0.040

The spectral solver needs the first-order form. Collocating $u_{tt} = \rho^{-1} \partial_x (E \partial_x u)$ directly is the obvious route and is unstable at the absorbing boundary: advancing u at the end nodes from u^{n-1} decouples them from the interior and the corner modes grow without bound within a few hundred steps. Splitting into velocity–stress makes the radiation condition an exact statement about the incoming Riemann invariant, which can be imposed as a projection at every Runge–Kutta stage. The difference is not accuracy but existence: the second-order form produced NaN, not a worse number.

Resampling can hide spectral convergence. Solvers choose their own time step from stability and their own spatial grid, so results are resampled onto a shared 201×101 evaluation grid. Doing that with linear interpolation is second-order accurate and puts a floor under the measured error at roughly 10^{-3} — which made the spectral solver look *algebraically* convergent and worse than finite differences. The fix is in two parts: round the step count up until the output times are hit exactly, removing time interpolation from the error budget entirely, and interpolate off the Chebyshev nodes barycentrically rather than linearly. After both, the spectral solver converges as it should. Any comparison of a spectral method against a low-order one on a shared output grid is exposed to this.

Continuation must carry the final iterate. We run continuation in 3 000-step blocks so training can stop once $w_{\min}$ reaches its target. Reloading the best validation checkpoint between blocks is not equivalent to continuing: frontier progress need not improve that scalar metric, so the next block can replay almost the same trajectory from the same parameters. The runner now passes each block’s final model directly to the next block and restores the best-loss checkpoint only once, before L-BFGS. Result files record this handoff mode, and every causal result reported in the paper was regenerated after the fix.

A note on the KAN implementation. The spline KAN used here is a direct Equinox implementation of a B-spline edge layer with a SiLU base branch (grid size G , spline order k), rather than a wrapper around an external package. It matches the pykan formulation but does not reproduce that package’s adaptive grid refinement, which would change the parameter shapes mid-run — neither the causal schedule nor L-BFGS tolerates that. The basis is verified to form a partition of unity and to admit the second derivatives the residual needs.

C Supporting tables

Two tables referenced from the body. Neither carries the main claim, so both sit here rather than against the page limit.

D Complete results

Table 6 lists every run in the study — 143 in total, across the main sweep and the weighting ablation. The tables in the body show slices chosen to make a point; this one hides nothing, so any cell of any other table can be checked against it.

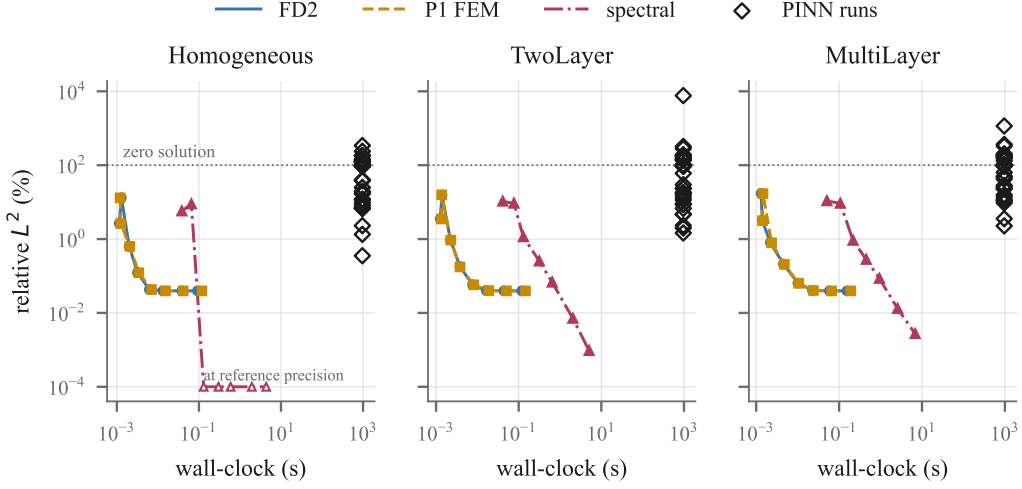


Figure 3: What accuracy costs. Wall-clock seconds against relative L^2 , both logarithmic. Connected markers are the classical refinement ladders (CPU, single core); hollow markers are the spectral solver agreeing with the reference to its own precision rather than a measured error. Diamonds are individual PINN runs at their measured training time on an H100. The dotted line at 100% is the score of predicting zero. The two families are separated by roughly four orders of magnitude in time and two to three in error.

Table 6: Every run in the study: 143 completed runs, one row each. Relative L^2 (%) against the finite-difference reference, taking the better of the Adam and L-BFGS checkpoints. *hard* is the constraint ansatz, *soft* the IC loss.

material	architecture	arm	IC	rel L^2 (%)
Homogeneous	ChebyshevKAN	base	hard	37.9
	ChebyshevKAN	base	soft	138
	ChebyshevKAN	lbfgs_only	hard	40.4
	ChebyshevKAN	no causal	hard	6.40
	ChebyshevKAN	no causal, no GN	hard	1.62
	ChebyshevKAN	rba	hard	11.4
	ChebyshevKAN	soap	hard	19.7
	ChebyshevKAN	soap + rba	hard	6.74
	FourierFeature	base	hard	2.30
	FourierFeature	base	soft	181
	FourierFeature	lbfgs_only	hard	6.93
	FourierFeature	no GradNorm	hard	79.3
	FourierFeature	no causal	hard	1.000
	FourierFeature	no causal, no GN	hard	0.302
	FourierFeature	rba	hard	1.34
	FourierFeature	soap	hard	339
	FourierFeature	soap + rba	hard	6.89
	FourierWavKAN	base	hard	120
	FourierWavKAN	base	soft	130
	FourierWavKAN	lbfgs_only	hard	16.8
	FourierWavKAN	no causal	hard	2.31
	FourierWavKAN	no causal, no GN	hard	0.624
	KAN (spline)	base	hard	123

continued on the next page

Table 6, continued

material	architecture	arm	IC	rel L^2 (%)
TwoLayer	KAN (spline)	base	soft	125
	KAN (spline)	lbfgs_only	hard	93.1
	KAN (spline)	no causal	hard	437
	KAN (spline)	no causal, no GN	hard	0.560
	PirateNet	base	hard	105
	PirateNet	base	soft	183
	PirateNet	lbfgs_only	hard	8.24
	PirateNet	no causal	hard	3.23
	PirateNet	no causal, no GN	hard	3.45
	PirateNet	rba	hard	17.8
	PirateNet	soap	hard	0.353
	VanillaPINN	base	hard	129
	VanillaPINN	base	soft	178
	VanillaPINN	lbfgs_only	hard	92.8
	VanillaPINN	no GradNorm	hard	12.9
	VanillaPINN	no causal	hard	7.37
	VanillaPINN	no causal, no GN	hard	1.16
	VanillaPINN	rba	hard	6.91
	VanillaPINN	soap	hard	12.1
	WavKAN	base	hard	24.9
	WavKAN	base	soft	145
	WavKAN	lbfgs_only	hard	96.8
	WavKAN	no causal	hard	21.3
	WavKAN	no causal, no GN	hard	2.70
	WavKAN	rba	hard	9.54
	WavKAN	soap	hard	238
	ChebyshevKAN	base	hard	13.2
	ChebyshevKAN	base	soft	135
	ChebyshevKAN	lbfgs_only	hard	17.3
	ChebyshevKAN	no causal	hard	10.4
	ChebyshevKAN	no causal, no GN	hard	3.16
	ChebyshevKAN	rba	hard	8.64
	ChebyshevKAN	soap	hard	23.2
	ChebyshevKAN	soap + rba	hard	4.62
	FourierFeature	base	hard	8.93
	FourierFeature	base	soft	177
	FourierFeature	lbfgs_only	hard	146
	FourierFeature	no causal	hard	1.01
	FourierFeature	no causal, no GN	hard	0.554
	FourierFeature	rba	hard	2.32
	FourierFeature	soap	hard	7648
	FourierFeature	soap + rba	hard	15.7
	FourierWavKAN	base	hard	18.2
	FourierWavKAN	base	soft	138
	FourierWavKAN	lbfgs_only	hard	29.7
	FourierWavKAN	no causal	hard	4.13
	FourierWavKAN	no causal, no GN	hard	0.923
	KAN (spline)	base	hard	95.9
	KAN (spline)	base	soft	321
	KAN (spline)	lbfgs_only	hard	93.7
	KAN (spline)	no causal	hard	112
	KAN (spline)	no causal, no GN	hard	0.836
	PirateNet	base	hard	104
	PirateNet	base	soft	193
	PirateNet	lbfgs_only	hard	6.75

continued on the next page

Table 6, continued

material	architecture	arm	IC	rel L^2 (%)
MultiLayer	PirateNet	no causal	hard	6.01
	PirateNet	no causal, no GN	hard	1.14
	PirateNet	rba	hard	1.99
	PirateNet	soap	hard	1.46
	VanillaPINN	base	hard	153
	VanillaPINN	base	soft	182
	VanillaPINN	lbfgs_only	hard	93.3
	VanillaPINN	no causal	hard	6.49
	VanillaPINN	no causal, no GN	hard	1.42
	VanillaPINN	rba	hard	4.74
	VanillaPINN	soap	hard	11.3
	WavKAN	base	hard	60.2
	WavKAN	base	soft	172
	WavKAN	lbfgs_only	hard	98.5
	WavKAN	no causal	hard	28.7
	WavKAN	no causal, no GN	hard	5.26
	WavKAN	rba	hard	23.7
	WavKAN	soap	hard	281
	ChebyshevKAN	base	hard	23.0
	ChebyshevKAN	base	soft	131
	ChebyshevKAN	lbfgs_only	hard	50.2
	ChebyshevKAN	no causal	hard	10.9
	ChebyshevKAN	no causal, no GN	hard	12.6
	ChebyshevKAN	rba	hard	25.7
	ChebyshevKAN	soap	hard	22.8
	ChebyshevKAN	soap + rba	hard	9.50
	FourierFeature	base	hard	11.9
	FourierFeature	base	soft	174
	FourierFeature	lbfgs_only	hard	10.9
	FourierFeature	no causal	hard	2.17
	FourierFeature	no causal, no GN	hard	1.51
	FourierFeature	rba	hard	2.27
	FourierFeature	soap	hard	328
	FourierFeature	soap + rba	hard	28.2
	FourierWavKAN	base	hard	13.9
	FourierWavKAN	base	soft	165
	FourierWavKAN	lbfgs_only	hard	42.1
	FourierWavKAN	no causal	hard	7.69
	FourierWavKAN	no causal, no GN	hard	5.55
	KAN (spline)	base	hard	97.2
	KAN (spline)	base	soft	364
	KAN (spline)	lbfgs_only	hard	94.7
	KAN (spline)	no causal	hard	207
	KAN (spline)	no causal, no GN	hard	5.71
	PirateNet	base	hard	105
	PirateNet	base	soft	203
	PirateNet	lbfgs_only	hard	11.0
	PirateNet	no causal	hard	4.18
	PirateNet	no causal, no GN	hard	2.76
	PirateNet	rba	hard	3.55
	PirateNet	soap	hard	1151
	VanillaPINN	base	hard	149
	VanillaPINN	base	soft	189
	VanillaPINN	lbfgs_only	hard	94.7
	VanillaPINN	no causal	hard	19.9

continued on the next page

Table 6, continued

material	architecture	arm	IC	rel L^2 (%)
	VanillaPINN	no causal, no GN	hard	16.2
	VanillaPINN	rba	hard	15.7
	VanillaPINN	soap	hard	96.0
	WavKAN	base	hard	63.9
	WavKAN	base	soft	165
	WavKAN	lbfgs_only	hard	99.8
	WavKAN	no causal	hard	23.0
	WavKAN	no causal, no GN	hard	16.5
	WavKAN	rba	hard	47.6
	WavKAN	soap	hard	191

Table 5: Variant training arms, hard-constraint IC, relative L^2 (%) at the better reported first- or second-order checkpoint. SOAP replaces Adam; RBA replaces causal weighting with residual-based attention; L-BFGS-only removes the first-order phase entirely.

material	architecture	base	SOAP	RBA	SOAP+RBA	lbfgs_only
Homogeneous	VanillaPINN	129	12.1	6.91	–	92.8
Homogeneous	FourierFeature	2.30	339	1.34	6.89	6.93
Homogeneous	PirateNet	105	0.353	17.8	–	8.24
Homogeneous	KAN (spline)	123	–	–	–	93.1
Homogeneous	WavKAN	24.9	238	9.54	–	96.8
Homogeneous	ChebyshevKAN	37.9	19.7	11.4	6.74	40.4
Homogeneous	FourierWavKAN	120	–	–	–	16.8
TwoLayer	VanillaPINN	153	11.3	4.74	–	93.3
TwoLayer	FourierFeature	8.93	7648	2.32	15.7	146
TwoLayer	PirateNet	104	1.46	1.99	–	6.75
TwoLayer	KAN (spline)	95.9	–	–	–	93.7
TwoLayer	WavKAN	60.2	281	23.7	–	98.5
TwoLayer	ChebyshevKAN	13.2	23.2	8.64	4.62	17.3
TwoLayer	FourierWavKAN	18.2	–	–	–	29.7
MultiLayer	VanillaPINN	149	96.0	15.7	–	94.7
MultiLayer	FourierFeature	11.9	328	2.27	28.2	10.9
MultiLayer	PirateNet	105	1151	3.55	–	11.0
MultiLayer	KAN (spline)	97.2	–	–	–	94.7
MultiLayer	WavKAN	63.9	191	47.6	–	99.8
MultiLayer	ChebyshevKAN	23.0	22.8	25.7	9.50	50.2
MultiLayer	FourierWavKAN	13.9	–	–	–	42.1

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Justification: Appendix A gives the exact commands, the run-id scheme, and the complete hyperparameter set, which is identical across every architecture and arm. Appendix B documents the three implementation choices (the velocity–stress formulation, the resampling scheme, and continuation-state handoff) that materially change the reported numbers.

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Justification: All code is in the accompanying repository, including the three classical solvers, the training runner with the ablation flags, and the scripts that regenerate every table, figure and inline number in this paper from the result files. There is no external dataset: the problem is a PDE with an analytic initial condition, so all data is generated by the solvers themselves.

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Justification: Section 4 gives the loss, both weighting schemes, the two initial-condition treatments and the two-phase optimiser schedule; Appendix A gives the collocation grid, step budgets, learning rate, causal and GradNorm settings, per-architecture sizes, and the model-selection criterion. Hyperparameters were not tuned per architecture, which is stated as a limitation.

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